



Department of Computer
Science and Engineering

CSE 1325 Project Documentation

PROJECT NAME

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PROJECT LOGO



Object oriented programming CSE 1325

Student declaration:

We declare that:

- *We understand what is meant by plagiarism*
- *The implication of plagiarism has been explained to me by our professor*
- *This assignment is all team own work and we have acknowledged any use of the published and unpublished works of other people.*

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1. Project Introduction

Describe briefly the main idea of the project.

The main idea of this project is to make internship opportunities more accessible to students and job seekers in fields with high competition, addressing the difficulty of finding relevant positions.

2. Project Description

A brief summary of the goals, the objectives, and the need for the project, including all its phases and processes involved to achieve them.

This program will allow a user to keep track of current and upcoming scholarships as well as internship opportunities. With every internship opportunity being tailored towards their degree and interest EX. aerospace engineers will get aerospace internship recommendations, while Music and Business majors will get their respective notifications on potential scholarship and internship opportunities.

The scholarship aspect allows you to search recent and upcoming scholarship, view information such as deadlines for applying and other eligibility criteria, it tracks your application in stages such as started, submitted, waiting for results and whether your application was accepted or declined. The internship aspect allows you to see a list of companies offering intern positions, their requirements, payment offers and whether it's a remote or in office job, you can mark some as interested and keep track of the application progress as well as receive interview updates, it keeps a record of which positions you've applied to, are waiting to hear back from, or have been accepted/rejected for.

There is potential for other features to come up as we go through with creating this project, but there is also potential for things to be cut from the feature list, this is currently a rough draft of how things should work out.

3. Project Architecture

Draw the architecture indicating all the different components that makes up this system and write a brief description of the architecture. An example of an architecture is shown below.

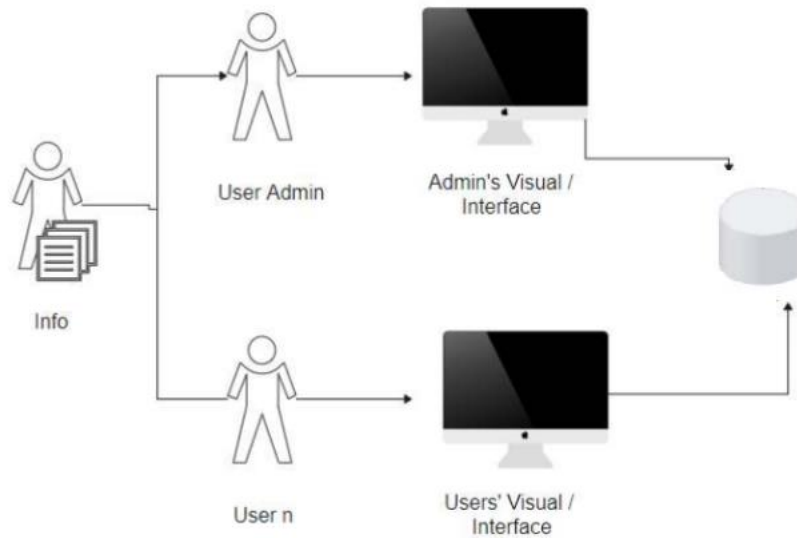


Figure 1 System architecture

4. UML Class Diagram

Generate class diagram that shows an overview of how all current classes and packages work together for your project. An example of a class diagram is shown below.

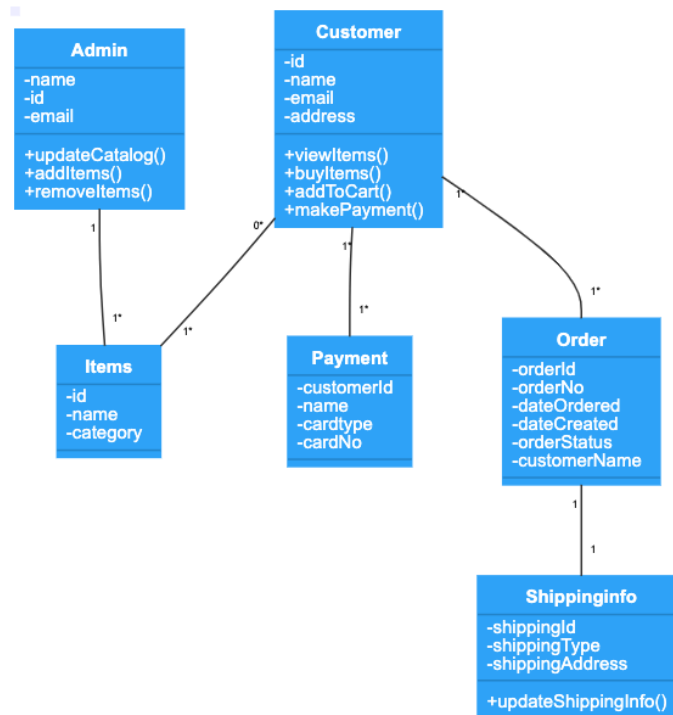


Figure 2 UML class diagram

5. OOP Concepts Used

Talk about all the different kinds of oop concepts used and describe them each in details. Also describe how you have used them to assemble your project.

project uses core OOP concepts to create a modular, organized system. **Classes** like Internship Tracker and InternshipDatabase separate user interaction from data management, while **encapsulation** protects internal data such as internship lists. **Enums** (Major, Year, Internship) enforce valid values and simplify data handling. **Methods** like getInternship() provide modular, reusable functionality, and **abstraction** hides the complexity of database operations from the main program. Features such as the account login system, enum-based internship database, and random internship selection demonstrate these principles in practice, making the program structured, maintainable, and user-friendly.

6. Code Description

Identify 3 key features of your system and present the sample code together with short descriptions to describe how they are used to implement specific parts of your system.

```
List<Internship> list = data.getDefault(major, Map.of())
                        .getDefault(year, List.of());
return list.get(ThreadLocalRandom.current().nextInt(list.size()));
```

Randomly selects one internship from a list.

Abstracted inside InternshipDatabase, so the main class doesn't care about selection logic.

Demonstrates **modularity, abstraction, and method usage**.

```
data.put(Major.COMPUTER_SCIENCE, new HashMap<>());
data.get(Major.COMPUTER_SCIENCE).put(Year.FIRST,
    List.of(Internship.INTRO_CODING));
```

Stores internship data using enums for Major and Year.

Ensures data integrity: only valid majors and years are allowed.

Demonstrates **encapsulation, enums, and OOP data modeling**.

```
BufferedReader reader = new BufferedReader(new FileReader(Username));
reader.readLine();
Pass_wrd_Conf = reader.readLine().trim();
```

Reads the stored password from a file and compares it to user input.

Implements security by limiting login attempts to 3.

Demonstrates **encapsulation and method usage** (input/output operations are abstracted).

7. System Testing

Provide screenshots of tests run on your completed system.

```
<terminated> Internship_Tracker [Java Application] C:\Users\HP\Downloads
Welcome to the Pathway+

Are you a returning user?(Yes/No):
no
Lets get an account started for you!
What username would you like to use?:
mcd
What is the password you would like to use?
mcd
Your account is now setup.
Run program again with your account information.
```

```
Internship_Tracker [Java Application] C:\Users\HP\Downloads\eclipse-comr
Welcome to the Pathway+

Are you a returning user?(Yes/No):
yes
Enter your usrxname:
mcd
Welcome back mcd,
Enter your password:mcd

Password confirmed

What would you like to do today?
1. Check on internship opportunities for you major:
2. Check on grants
3. Check current internships in grants on your account.

1
What is your major
CS-Computer Science
ME - Mechanical Engineering
BE - Biomedical Engineering
AE - Aerospace Engineering
NU - Nursing
CE - Civil Engineering
BIO - Biology
Select your choice:
cs
What year are you in?:
```

```
CS
What year are you in?:
2

Software Testing Internship:

The internship involves learning and executing various testing methodologies. Co
Interns gain practical experience with essential testing tools (e.g., Selenium f

Are you interested in accepting this position?

yes

Internship Accepted!
```

8. Group members

Explain the individual responsibility of each team member inside the project.

Mc-Daniels – created the internship selection using info about the user, helped various members with their parts

Samuel – Created the account set up and internship tracking and distributed task to team members

Jean – handled grants for scholarships

9. Conclusion and future works

It is concluded that the main objective of the application is to help computer science students understand the basics of Java. By browsing through the application and looking at the code for each graphical interpretation, students should be able to easily understand the implementation. Future work could involve adding more categories which are more detailed and have additional items. This part of the documentation is reserved for conclusion and future possible improvements on the project itself.